Phase-change materials for rewriteable data storage

Phase-change materials are some of the most promising materials for data-storage applications. They are already used in rewriteable optical data storage and offer great potential as an emerging non-volatile electronic memory. This review looks at the unique property combination that characterizes phase-change materials. The crystalline state often shows an octahedral-like atomic arrangement, frequently accompanied by pronounced lattice distortions and huge vacancy concentrations. This can be attributed to the chemical bonding in phase-change alloys, which is promoted by *p*-orbitals. From this insight, phase-change alloys with desired properties can be designed. This is demonstrated for the optical properties of phase-change alloys, in particular the contrast between the amorphous and crystalline states. The origin of the fast crystallization kinetics is also discussed.

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The ability to store information has been an important aspect of human development. Ancient cave paintings, hieroglyphs on pyramids in Egypt, the eighth-century archival scrolls of the Shosoin treasury in Nara, Japan, or medieval book printing remind us of the human desire to preserve knowledge and information for future generations. The past century has seen the development of a plethora of new techniques for storing huge amounts of data, one of the most successful being optical data storage[1]. Whereas the first optical storage products only allowed reading of the encoded information, nowadays both recordable and rewriteable storage media have been developed.

Initially two different classes of rewriteable optical storage media have been commercialized, namely magneto-optical and phase-change media. In recent years, however, most rewriteable products have used phase-change materials. This success is based on the discovery[2] of alloys containing elements such as Ge, Sb and Te, and the tailoring of phase-change media to meet DVD specifications[3]. Here, our goal is to explain the principle of phase-change recording and to describe which materials fulfil the unique requirements that characterize rewriteable data storage. To do so, we present and discuss the atomistic mechanisms used together with the issues that have not yet been resolved.

The relentless pursuit of material optimization and the drive to understand the underlying material characteristics have led to the development of three generations of optical storage products—compact disks (CD), digital versatile disks (DVD), and Blu-ray disks (BD) or high-definition digital versatile disks (HD DVD), respectively. We present these briefly, before discussing different approaches to developing a fourth-generation product and looking at several of the open issues. In recent years phase-change materials have also become of interest for developing non-volatile memories with a number of attractive features[4–6], and we consider the potential of this storage concept as well as the unique material properties required.

In applications as a non-volatile memory, the pronounced difference in electrical resistivity is used. The amorphous state has a high resistance. Applying a long voltage pulse (set pulse) locally heats the amorphous region and leads to recrystallization. Applying a higher voltage pulse (reset pulse) to the crystalline state with low resistance leads to local melting and the formation of an amorphous region on rapid quenching.

The storage concept used in phase-change materials is sketched and explained in Fig. 1. There are many materials that can be melt-quenched to form an amorphous state, but very few materials also possess a pronounced difference in optical properties between the amorphous and crystalline state. Hence only a very few materials have the unique property combination found in phase-change materials used for optical storage. The pronounced optical contrast indicates that the atomic arrangement differs considerably between the amorphous and crystalline states. Nevertheless the atomic rearrangement required on recrystallization proceeds on a very fast timescale of ten to one hundred nanoseconds[2,7]. Table 1 summarizes the combination of properties that characterizes phase-change materials. It is interesting to note that the required contrast in electrical conduction is much easier to meet, so there might be many more candidates for non-volatile memory than for rewriteable optical storage. We return to this point later.

It is evident that most materials will fail to meet one or other of the requirements listed in Table 1. The question thus arises of how we can systematically identify materials that can fulfil the conditions outlined above. Suitable materials for rewriteable optical storage that have been identified in the past few decades are shown in the schematic phase diagram of Fig. 2.

The first materials used were good glass formers such as Te-based eutectic alloys, represented by $Te_{85}Ge_{15}$, doped with elements such as Sb, S and P (ref. 4). Although these materials already showed electrical switching that could be used for electronic storage, the time for crystallization was of the order of microseconds, partly because the first alloys did not crystallize

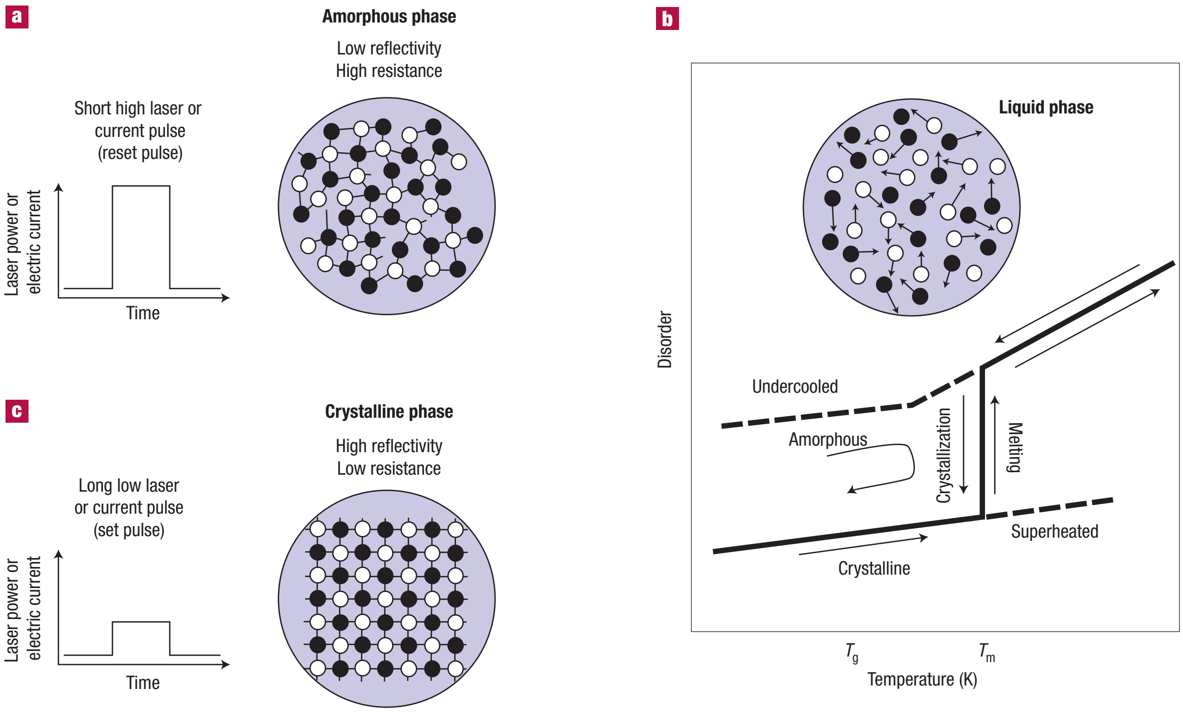

Figure 1 Rewriteable optical data storage using phase-change materials. **a**, A short pulse of a focused, high-intensity laser beam locally heats the phase-change material above its melting temperature. **b**, Rapidly cooling the alloy at rates higher than 10^9 K s^{-1} quenches the liquid-like state into a disordered, amorphous phase. This amorphous state has different optical properties from the surrounding crystalline state, so detecting amorphous regions is straightforward with a low-intensity laser beam. **c**, To erase the stored information a laser pulse with intermediate power is used. The laser locally heats the phase-change film above the crystallization temperature. At temperatures above T_g the atoms become increasingly mobile and can revert to the energetically favourable crystalline state, erasing the recorded information.

in a single-phase material. The first materials to show fast recrystallization and good optical contrast were GeTe (ref. 7) and $Ge_{11}Te_{60}Sn_4Au_{25}$ (refs 8,9). This triggered the discovery of pseudo-binary alloys along the GeTe–Sb_2Te_3 tie line, such as $Ge_1Sb_4Te_7$, $Ge_1Sb_2Te_4$ and $Ge_2Sb_2Te_5$ (refs 2,10). Nowadays $Ge_2Sb_2Te_5$ and related materials such as GeSbTeN, GeSnSbTe, GeBiSbTe, GeBiTe and GeInSbTe (refs 11–14) have been tried, some of which are frequently used in commercial products. A second material family of doped Sb_2Te alloys was found later, in which dopants such as Ag, In and Ge have often been used[15–17]. A composition close to $Ag_5In_5Sb_{60}Te_{30}$ (AIST) is frequently used in rewriteable optical storage media such as DVD-RW and DVD+RW[18]. In recent years a third material family has been found[19,20] that uses Ge-doped Sb.

The next section addresses the atomic arrangement that characterizes the crystal structure of the (metastable) crystalline state. Should all phase-change alloys show a similar atomic arrangement, then it would be possible to use the structure of new materials to identify those that might show phase-change properties, greatly helping in the search for new alloys.

STRUCTURE OF PHASE-CHANGE ALLOYS

CRYSTAL STRUCTURES

Figure 3a depicts the structure of a typical GeSbTe alloy. As can be seen, the Te atoms (light blue) occupy one lattice site, and Ge and Sb atoms (dark blue) randomly occupy the second lattice site of an atomic arrangement that closely resembles the rocksalt structure. There are a few noteworthy differences from an ideal rocksalt structure. In a material such as $Ge_1Sb_2Te_4$ or $Ge_2Sb_2Te_5$ there are a considerable number of vacancies[21,24] as schematically depicted in Fig. 3a. For $Ge_1Sb_2Te_4$, for example, there are 25% vacancies on the Ge/Sb site, whereas the Te site is fully occupied. In addition to large vacancy concentrations, the systems studied so far are characterized by both static and temperature-induced displacements or distortions[25]. Convincing evidence for pronounced distortions has also been found in extended X-ray absorption fine structure (EXAFS) experiments[26]. Not only did these measurements reveal a striking difference in short-range order between the amorphous and crystalline states, but the spectra for the crystalline state recorded at room temperature were characterized by pronounced local distortions. We will return to the origin of such distortions later.

Figure 3 also shows the atomic arrangement for two other typical classes of materials. In Fig. 3b, the crystalline structure of $Ge_4Te_{60}Sn_{11}Au_{25}$ is depicted[22]. This material crystallizes in a simple cubic phase. In this case all atoms statistically occupy the lattice sites. No evidence for local distortions (such as, for example, in $Ge_2Sb_2Te_5$ or $Ge_1Sb_2Te_4$) is found. Finally a third class of materials including AIST alloys (Fig. 3c) is found, which shows a slightly distorted simple cubic structure, that is, a statistical distribution of atoms over the lattice sites, yet with small distortions. It is interesting to note that this phase shows a large atomic displacement at high temperature where the structure becomes almost perfectly cubic[27].

In summary, the characteristic feature of all phase-change alloys currently used for optical storage is an octahedral-like atomic arrangement. Depending on the specific alloy, the material also possibly possesses local distortions and different degrees of chemical order regarding the occupation of lattice sites as well as the appearance of vacancies. This structural similarity is indicative of a characteristic type of bond that is encountered in phase-change alloys.

Table 1 Crucial properties of phase-change alloys

Required property of PC material	Specification
High-speed phase transition	Induced by nanosecond laser or voltage pulse
Long thermal stability of amorphous state	At least several decades at room temperature
Large optical change between the two states (for rewriteable optical storage)	Considerable difference in refractive index or absorption coefficient
Large resistance change between the states (for non-volatile electronic storage)	Natural consequence of the transformation from amorphous to crystalline state
Large cycle number of reversible transitions	More than 100,000 cycles with stable composition
High chemical stability	High water-resistivity

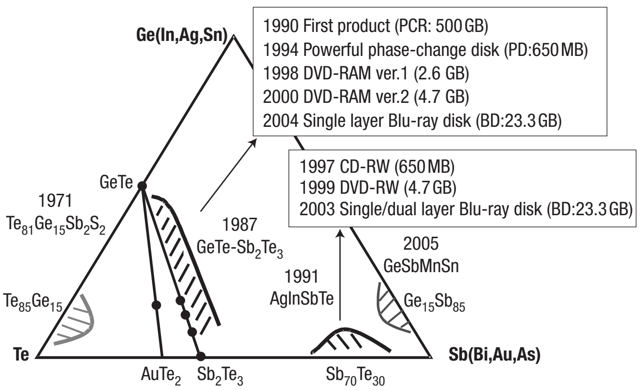

Figure 2 Ternary phase diagram depicting different phase-change alloys, their year of discovery as a phase-change alloy and their use in different optical storage products.

STRUCTURES OF THE AMORPHOUS PHASE

It is usually assumed that the amorphous state has an atomic arrangement that closely resembles the crystalline state locally but lacks its long-range order[28]. This assumption is plausible and was used as early as 1932 by Zachariasen when he developed a model for glass-forming ability, which led to the random covalent network theory[29]. The underlying idea of covalent network theory is a clear hierarchy of bonds: that is, that some bonds are stronger than others. In a typical covalent structure, the nearest-neighbour spacing is almost constant, but some variation of bond angles is found. Hence, even though the amorphous state lacks the long-range order of the crystalline state, there still are clear rules on how to arrange the atoms locally. The structure of the glasses can thus usually be predicted by local energy minimization. Random covalent network theory has been extended in the past few decades beyond a mean-field description[30–32]. It can now be used to address a variety of observations related to glasses and glass formation including the width of the intermediate phase that separates floppy and stressed-rigid glasses.

Although these scientific successes demonstrate the power to predict glass properties for many materials, a straightforward application of this theory to phase-change alloys has not yet been demonstrated. It seems that a solid understanding of the bonding mechanism is a prerequisite. This task is complicated for phase-change alloys, for which there is strong experimental evidence that the local atomic arrangements in the amorphous and crystalline state differ considerably[26]. The EXAFS spectra of the amorphous state could be reproduced by assuming a tetrahedral atomic arrangement of the Ge atoms, whereas the same atoms have an octahedral-like atomic arrangement in the crystalline phase. Kolobov *et al.*[26] have termed the transition between the amorphous and the crystalline states an "umbrella flip" and attributed the fast crystallization speed to the similarity in the atomic arrangement. This finding might provide a first clue to how to explain optical contrast. At the same time, the speed of crystallization cannot be directly related to the local atomic rearrangement. Furthermore, as the analysis of EXAFS spectra for ternary systems is a formidable task, a number of studies have been made to confirm or refute the conclusions drawn in ref. 26.

In a theoretical investigation of $Ge_1Sb_2Te_4$, a phase-change alloy that closely resembles $Ge_2Sb_2Te_5$ in terms of structure and properties, different atomic arrangements were studied[33]. These computations demonstrated unequivocally that in $Ge_1Sb_2Te_4$ different arrangements with very similar energy are possible. In particular, the structure with the lowest energy leading to a cubic crystalline phase is characterized by an octahedral atomic arrangement, yet with pronounced local distortions. The second atomic arrangement, which differs by a very modest 30 meV per atom, is the spinel structure. In this structure every Ge atom has four nearest Te neighbours (tetrahedral arrangement), and the Sb and Te atoms have similar positions to those in the rocksalt structure. Hence the local atomic arrangement in the amorphous structure could resemble the spinel structure. A spinel structure is also frequently observed for the lighter chalcogenides such as selenides, sulphides and oxides.

Subsequent calculations have also found evidence that a tetrahedral atomic arrangement of the Ge atoms as in the spinel structure could account for the apparent increase of the bandgap in the amorphous state[33,34]. The existence of two different atomic arrangements with similar energy is further evidence for a rather complex energy landscape[35], a concept that is becoming increasingly important in describing glass properties. At the same time, the calculations are based on a model using just 64 atomic positions, and thus not ideally suited to describe the absence of long-range order, which is also characteristic for the amorphous state.

Based on a recent EXAFS analysis, Baker and co-workers[36] find evidence for a significant number of Ge–Ge bonds in $Ge_2Sb_2Te_5$. Subsequently, using constraint theory, they concluded that in the amorphous state of $Ge_2Sb_2Te_5$ an ideal network structure is formed. Indeed, evidence for homopolar bonds has also been found in several studies of GeTe, but other recent studies (discussed later) do not find evidence for such bonds in GeSbTe alloys. The conclusion that GeSbTe-based phase-change alloys are good glass formers has been put forward to explain why phase-change alloys show good cyclability and can be switched very rapidly between the amorphous and crystalline state[36]. Good glass formers, however, are characterized by long crystallization times; a property neither found nor desired in phase-change alloys.

Other studies of the amorphous state have used X-ray diffraction, which offers a large range of wavevector transfers and hence data that can make precise modelling of the local structure possible. From such a study, Kohara *et al.*[37] conclude that amorphous $Ge_2Sb_2Te_5$ is characterized by even-folded ring structures, whereas amorphous GeTe has both even- and odd-numbered rings. This finding is schematically depicted in Fig. 4. Whereas odd-numbered rings are evidence for homopolar bonds such as Ge–Ge bonds, the even-numbered rings are consistent with a strong preference for heteropolar bonds and hence the absence of Ge–Ge bonds in $Ge_2Sb_2Te_5$. The corresponding calculations reveal that microscopically many relics of the crystal structure are preserved in the amorphous structure, such as even-numbered rings and bond angles that centre around 90°, although the total pair correlation functions for the amorphous and crystalline state are rather different[37].

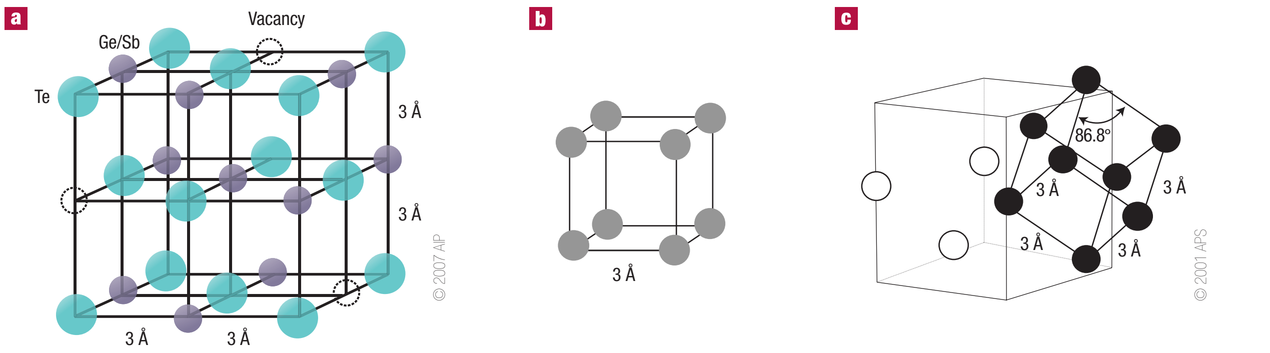

Figure 3 Crystal structures. **a**, Schematic image of the crystal structure of the rock-salt-like phase of, for example, $Ge_2Sb_2Te_5$ or $Ge_1Sb_2Te_4$. Te atoms occupy one sublattice of the crystal, and Ge atoms, Sb atoms and vacancies randomly occupy the second sublattice. The typical nearest-neighbour spacing is close to 3 Å. Local distortions are not depicted. Reprinted with permission from ref. 21. **b**,The simple cubic structure found, for example, in $Te_{60}Ge_4Sn_{11}Au_{25}$. The atoms are statistically distributed over the sites of an undistorted lattice[22]. **c**, The statistical distribution of atoms over the sites of a rhombic lattice[23]. This atomic arrangement is found in, for example, $Ag_5In_5Sb_{60}Te_{30}$ as a high-temperature phase. Figure reprinted with permission from ref. 27.

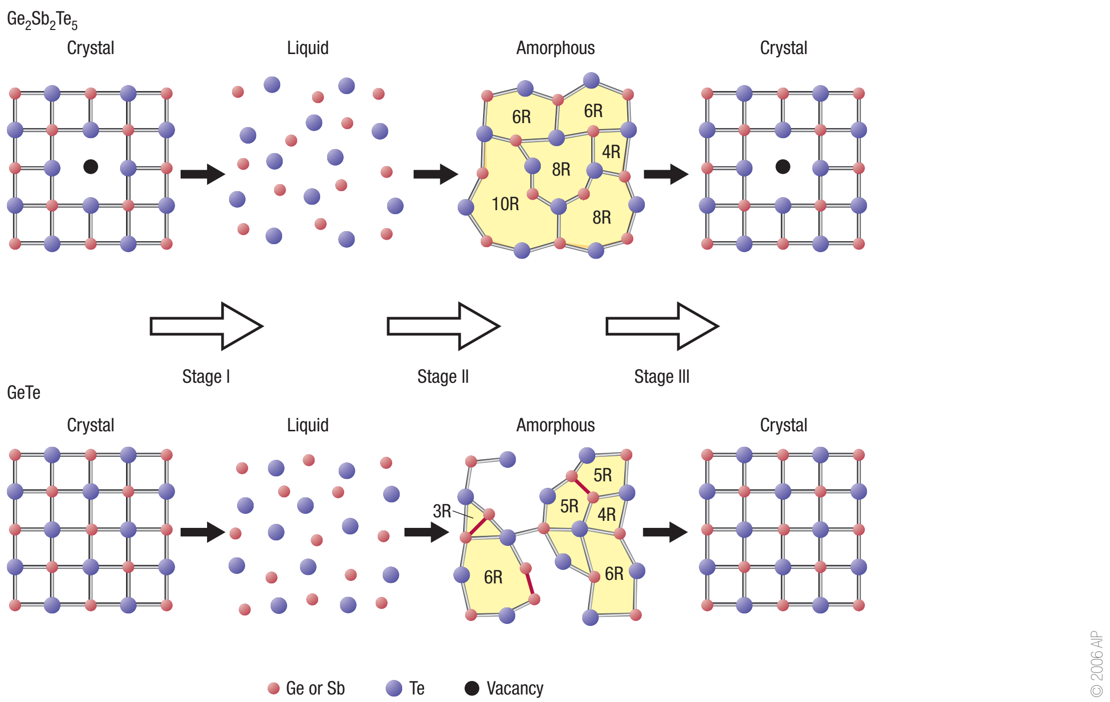

Figure 4 Schematic presentation of the possible ring structure transformation in the phase changes crystal–liquid–amorphous (record) and amorphous–crystal (erase) in $Ge_2Sb_2Te_5$ and GeTe. The red bonds represent the Ge–Ge bond. Stage I and II: recording process in optical storage; stage III: erasing process in optical storage. Reprinted with permission from ref. 37.

CHARACTERISTIC PROPERTIES OF PHASE-CHANGE MATERIALS

REMARKABLE CRYSTALLIZATION KINETICS

The two key properties that characterize phase-change materials are their remarkable crystallization kinetics and their contrast between the amorphous and crystalline phases. We briefly outline these aspects and present some concepts that can explain some recent observations[38–40].

The schematic presentation of data storage in phase-change media shown in Fig. 1 implicitly contains one important aspect of the crystallization kinetics in phase-change alloys, which is related to the span of timescales in this class of materials. The length of amorphous regions in optical disks corresponds to the encoded information. Therefore, it is crucial for data integrity that this length does not decrease by partial recrystallization while the media is stored at room temperature or while neighbouring tracks are (over)written. At room temperature, the amorphous region should be stable for at least 10 years. The information is erased by a pulsed laser, which heats the amorphous region above the glass-transition temperature. The increased atomic mobility allows fast recrystallization at elevated temperature within several tens of nanoseconds. Hence the amorphous region must not recrystallize at, say, 30 °C within 10 years

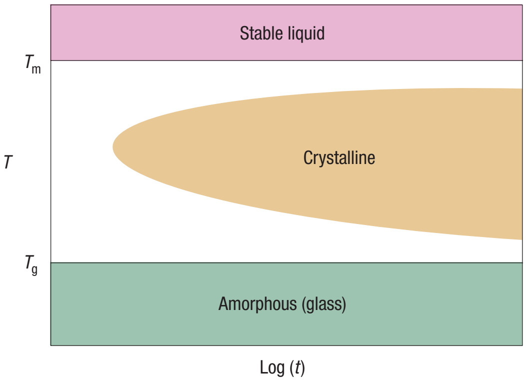

Figure 5 Time–temperature–transformation diagram of an undercooled liquid. t = time. The grey region depicts schematically that the minimum time for crystallization strongly depends on temperature.

(3×10^8 s), but should recrystallize somewhere in the range between 250 and 550 °C within 10^{-8} s. In other words, the timescale needs to change by roughly 16–17 orders of magnitude on just increasing the temperature by a few hundred kelvin. To understand which materials possess such a large range of crystallization speeds, a brief summary of crystallization theory is required.

In the following we focus on the fast recrystallization process that takes place below the melting temperature (T_m) but above the glass-transition temperature (T_g). We do not discuss any undesired recrystallization of amorphous marks below the glass transition temperature. A schematic plot of the crystallization speed between T_g and T_m is depicted in Fig. 5. As can be seen from this plot, crystallization of a liquid phase is thermodynamically allowed once the material is cooled below the melting temperature (undercooled liquid). The free-energy difference between liquid and crystal (driving force) favours the crystalline state. It vanishes at T_m, where the liquid and the crystalline phase coexist, but increases on cooling below T_m, so the driving force to form crystalline nuclei increases with decreasing temperature. At the same time atomic mobilities rapidly decrease. Recrystallization is therefore slow close to the melting temperature, as the high atomic mobilities are over-compensated by the marginal driving force for nucleation. At temperatures slightly above T_g, on the other hand, the driving force for nucleation is high but now atomic mobilities are low, again resulting in slow recrystallization. The highest crystallization speeds are encountered at intermediate temperatures between T_g and T_m, where a good compromise between atomic mobilities and the driving force enables a fast transition. Turnbull has shown that the reduced glass-transition temperature T_{rg} (= T_g/T_m), which varies between 0.4 and 0.85 for any glass former, is inversely related to the nucleation rate in a variety of materials[41]. Materials with a value of T_{rg} larger than around 0.7 feature low nucleation rates and can thus be quenched into a glassy state without crystallization, even during slow cooling. They are therefore known as easy glass formers. Until recently, values of T_g for phase-change materials had not been determined, but a modified differential scanning calorimetry experiment has now allowed T_g to be unequivocally identified[40]. It was shown that T_g is close to the crystallization temperature and that T_{rg} varies between 0.45 and 0.55 for different phase-change materials. These relative low values for T_{rg} imply that the materials are marginal glass formers and hence recrystallize very rapidly at elevated temperatures. However, T_{rg} values alone are too simple a parameter to describe crystallization kinetics exclusively[42]. An important parameter not discussed so far is the interfacial energy between the amorphous matrix and the crystalline nucleus, which has recently been estimated from undercooling experiments[42]. These experiments have provided strong evidence that the interfacial energies in phase-change alloys are rather low, explaining the small size of critical nuclei observed in electron microscope images[43]. The latter result is important because it ensures that phase-change recording should be feasible even for feature sizes below 20 nm (ref. 44). For such small dimensions interfacial effects become increasingly important, and can lead to heterogeneous rather than homogeneous nucleation. Even in many of the studies performed up to now on thin films, strong evidence for heterogeneous nucleation is found[45]. Hence the role of the interface is crucial in achieving the desired crystallization characteristics.

A second important point should be added when discussing the potential of phase-change recording. It has been frequently observed that recrystallization of an amorphous bit is considerably faster than the first crystallization of an as-deposited amorphous film. Two different processes may be responsible[46,47]. If recrystallization proceeds through growth from the crystalline rim, then this avoids the nucleation that is necessary for the first crystallization of an amorphous material. But even for those materials in which nucleation dominates the recrystallization process of an amorphous mark in a crystalline matrix, the melt-quenched state seems to contain some subcritical crystalline nuclei, which aid recrystallization and reduce the incubation time for producing these critical nuclei.

Over the past decade, efficient material optimization schemes have been developed in industry (Matsushita, Philips, Ricoh) to identify alloys with high optical contrast and fast recrystallization at elevated temperatures. Progress has also been made in identifying the underlying origin of optical contrast and recrystallization speed. A deeper insight would provide an alternative optimization scheme using quantum mechanical computations for material design.

PROPERTY CONTRAST BETWEEN AMORPHOUS AND CRYSTALLINE STATES

For typical covalent semiconductors such as GaAs, there is only a modest difference in the dielectric function ε of the amorphous and crystalline states. The most obvious difference between the two phases is the reduction in the amplitude of absorption in the amorphous phase which leads to a decrease in the peak maximum for ε_2. Furthermore, the width of this peak is broader for this phase, as expected because of smearing of electronic states and the formation of tail states in amorphous materials[48]. However, for a typical phase-change alloy such as $Ge_2Sb_2Te_5$, a much more pronounced contrast is observed. As the density between the amorphous and crystalline phase also shows a pronounced changed of 5–10% (ref. 49), it has been suggested that this density change is responsible for the large optical contrast. More specifically, recent studies[26,33] have suggested that it is the different atomic arrangement that is responsible for the different optical properties.

The lack of an established model for the amorphous phase and the difficulties in calculating optical properties from first principles also impede the design of phase-change materials, and in particular their optical contrast, based on theoretical concepts. In the past, these alloys have mainly been optimized by reverting to ingenious empirical strategies. The introduction of different generations of storage families — that is, the development of the first, second and third generation of optical storage devices — has created the need to optimize optical contrast for decreasing laser wavelength. A particular challenge has been the identification of phase-change alloys with sufficient contrast for wavelengths around 405 nm (third-generation storage products). Figure 6 shows how the contrast depends on composition for GeSbTe alloys and increases with increasing Ge content. The $Ge_2Sb_2Te_5$ compound that showed good contrast in the infrared and red spectral range is not ideal for shorter wavelengths. Alloys such as $Ge_8Sb_2Te_{11}$ or those with more Ge content, possess

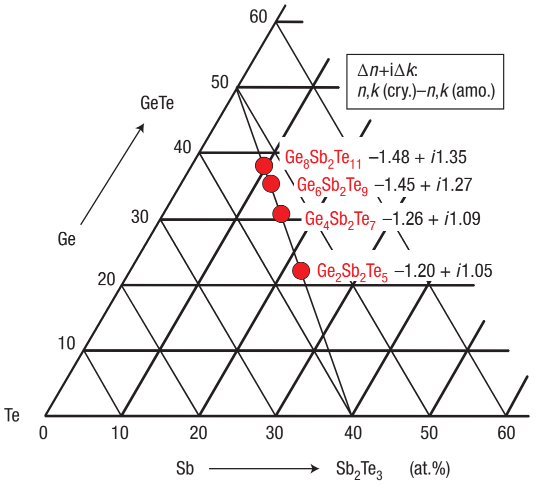

Figure 6 Change of refractive index (*n*) and absorption coefficient (*k*) with stoichiometry for a wavelength of 405 nm. The contrast increases with increasing Ge content along the pseudo-binary line[50].

Table 2 Some successful and unsuccessful phase-change materials, their structures and their average number of valence electrons (data from ref. 6).

Material	Structure	Average number of valence electrons
Successful samples*		
GeTe	Rocksalt	5
$Ge_1Sb_2Te_4$	Rocksalt (metastable)	4.75
$Ge_2Sb_2Te_5$	Rocksalt (metastable)	4.8
$Ge_4Sb_1Te_5$	Rocksalt	5.1
In_3SbTe_2	Rocksalt	4.33
$AgSbTe_2$	Rocksalt	4.5
$AuSbTe_2$	Rocksalt	4.5
$Au_{25}Ge_4Sn_{11}Te_{60}$	Cubic	4.45
$Ag_3In_4Sb_{76}Te_{17}$	Cubic	4.93
$Ag_{5.5}In_{6.5}Sb_{59}Te_{29}$	Cubic-like/hexagonal	4.94
Unsuccessful samples		
$AgInTe_2$	Chalcopyrite	4
$AuInTe_2$	Chalcopyrite	4

*Success is defined as the existence of a significant difference in the dielectric function of the amorphous and crystalline state ensuring optical contrast. Note that $Ge_1Sb_2Te_4$ and $Ge_2Sb_2Te_5$ each have one vacancy per unit cell in the cubic structure.

the required optical contrast for wavelengths around 405 nm and are used in third-generation storage media such as BD disks.

At present, theoretical approaches to reproduce or, better yet, to predict or design optical contrast are extremely sparse. In the first *ab initio* calculation of optical properties in phase-change alloys[34], it was demonstrated that the optical contrast experimentally observed in, for example, $Ge_1Sb_2Te_4$ could be qualitatively reproduced if a considerable fraction of Ge atoms in the amorphous state occupied tetrahedral positions. The following results of this investigation are particularly noteworthy.

First, the assumption that all Ge atoms in the amorphous phase occupied tetrahedral sites produced spectra that are not consistent with the experimental data. This conclusion emphasizes how important it is to develop an unambiguous model of the amorphous state. The experimental data could be reproduced, however, if 50% of the Ge atoms occupied tetrahedral sites, and the remaining Ge atoms occupied distorted octahedral sites. This would lead to an average coordination number between 3 and 4, consistent with the reverse Monte Carlo (RMC) calculations[37].

Second, the contrast in the optical properties could be analysed in detail in the quantum mechanical calculations by using Fermi's golden rule[51]. The corresponding computations revealed that it is the change in the matrix elements for the optical transition between the initial and final states that accounts for most of the observed contrast. This is an important conclusion as it is usually assumed that the change in the density of states, not the change in the matrix elements for optical transitions, explains the different optical properties in the amorphous and crystalline states of covalent semiconductors such as GaAs (ref. 52). This is clearly not the case for GeTe and $Ge_1Sb_2Te_4$. Finally, the computations showed that the lattice distortions in phase-change alloys decrease the optical contrast[34]. These distortions are particularly pronounced for high vacancy concentrations and low Ge content[53]. Moving to alloys with low concentrations of intrinsic vacancies and high Ge content should therefore increase the optical contrast. This is in good agreement with the trend in the experimental data sketched in Fig. 6. It seems likely that quantum mechanical calculations will be used to optimize or design phase-change materials in the near future. An improved knowledge and understanding of the amorphous state would make this approach even more powerful.

RELATIONSHIP BETWEEN STRUCTURE AND BONDING

In the section on characteristic structures, we noted that in all phase-change alloys identified up to now the atoms show an octahedral-like arrangement, often accompanied by local distortions. Another characteristic feature is the high intrinsic vacancy concentration found in many phase-change alloys based on GeSbTe. Density functional theory (DFT) has recently been used to understand the origin of these properties. These computations show that the removal of Ge or Sb atoms from a $Ge_2Sb_2Te_4$ crystal with rocksalt structure lowers the energy of the crystal[53]. Such a finding is very unusual because removal of atoms requires the breaking of bonds and typically is accompanied by high vacancy formation energies. This is clearly not the case for $Ge_2Sb_2Te_4$. Quantum chemical calculations have revealed the origin of this effect. They showed that in $Ge_2Sb_2Te_4$ a considerable fraction of antibonding states is occupied. Removal of atoms reduces the average number of valence electrons and hence the occupation of antibonding states, leading to a more stable compound. It was also demonstrated that distortions lead to an additional strong decrease in energy, stabilizing the alloys further[53]. The octahedral-like atomic arrangement has been attributed to the formation of *p*-bonds between the atoms. Such *p*-bonds are prone to local distortions, which are frequently denoted as Peierls distortions[53–56].

The question should therefore be raised: under which circumstances do *p*-bonding and the accompanying octahedral atomic arrangement prevail over the sp^3-bonding encountered in most semiconductors? For alloys with a similar concentration of chalcogen and non-chalcogen atoms, DFT has been used to determine the energy difference between sp^3-bonded phases with tetrahedral atomic arrangement and *p*-bonded systems with octahedral atomic arrangement[6]. A systematic variation of the most stable crystal structure with the average number of valence electrons was observed. All alloys with an octahedral atomic arrangement were found to have more than four valence electrons whereas alloys with four valence electrons were characterized by a tetrahedral atomic arrangement. Interesting enough, as noted above, all successful phase-change alloys found so far are characterized by an octahedral-like atomic arrangement (Table 2). Hence the search for new phase-change alloys can be simplified considerably by scanning the class of group 16 and group 15 alloys that have octahedral-like atomic arrangements and a suitable value of T_g/T_m around 0.5. These rules already define stringent selection criteria that apparently must be met. As can be seen from Table 2, the search for alloys with

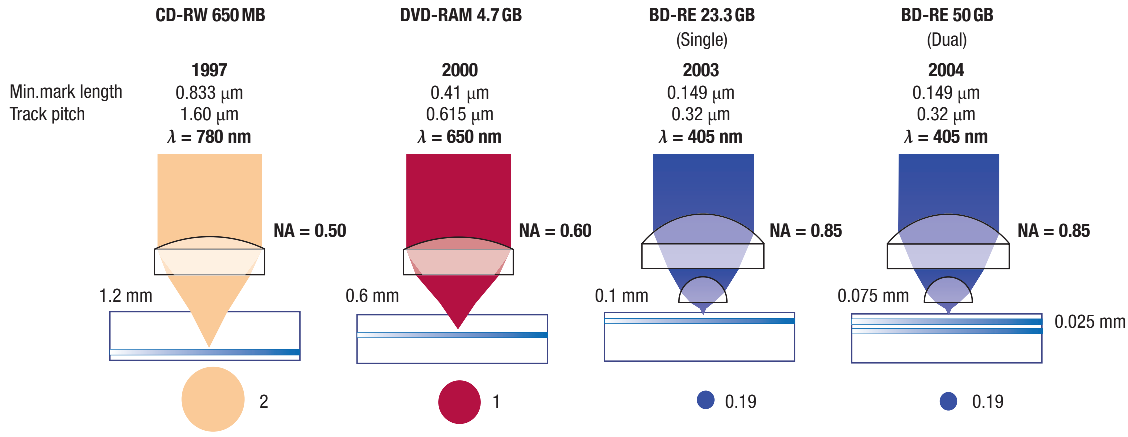

Figure 7 Comparison of optics, recording densities, recording capacities and disk structures used for CD, DVD and BD. Changes in the lens system are shown in the upper portion; the disk structure is depicted beneath. The change in beam cross-section is displayed at the bottom. From CD to DVD and DVD to BD, the effective laser beam cross-section has been reduced by a factor of 2 and 5, respectively. The recording density increased by a factor of 7 going from CD to DVD. Going from DVD to the dual-layer BD another increase by a factor of 11 has been accomplished.

octahedral-like atomic arrangement can be further simplified if the average number of valence electrons is used, providing a particularly straightforward selection criterion. But the last rule has to be applied with care. It holds if both chalcogen and non-chalcogen constituents are present in about equal concentrations, but cannot be applied in cases with a considerable excess of one type of atom, as in $GeSe_2$. In this case the suitability of a given alloy has to be derived from its crystal structure.

Having addressed the properties of phase-change alloys and the identification of new materials suitable for rewriteable storage, we turn now to the different generations of rewriteable storage solutions that are made possible by phase-change alloys.

Figure 7 shows the progress of the phase-change optical disk in the past decade. The recording density has been increased by roughly a factor of a hundred. Simultaneously the data-transfer rate was increased by a factor of a thousand. This raises the tantalizing question of what the next generation of optical storage products will look like. Interestingly enough, there is strong reason to believe that the fourth generation will look very different from any predecessor.

In past decades, progress was made by reducing the wavelength of the laser used, from infrared to red and from red to blue-violet, increasing the numerical aperture of the lens used, from 0.5 to 0.6 and from 0.6 to 0.85, doubling the recording layers from single to dual, and making use of improved software algorithms. Further reduction in laser wavelength faces fundamental challenges. Not only is there a lack of suitable laser diodes with a wavelength in the ultraviolet range, but the substrates now used (for example polycarbonate but also glass) become absorbing in the ultraviolet. As a consequence it has to be expected that the road of evolutionary development can no longer be followed. Instead it will be necessary to break new ground.

To maintain the pace of product development, a number of competing technologies are being considered. At present it is not clear which approach will prove superior, so only two concepts are briefly sketched. One is the near-field approach, in which a lens with a large numerical aperture is brought very close to the storage medium. At distances $d < \lambda/2$, the evanescent wave still has components at the wavelength inside the lens, that is, with $\lambda' = \lambda/n$, where n is the refractive index of the material. These components allow much higher spatial resolution and ideally could increase the storage density by $1/n^2$. Working prototypes have been announced by several companies (including Philips and Sony). Because the proof of concept has already been achieved, market introduction is expected, but probably not in the near future. The main challenge for system reliability is the close spacing between the lens and the storage medium. This situation resembles that encountered in magnetic recording where a magnetic head is held close to the recording disk. As far as the phase-change alloy itself is concerned, no major challenges are envisaged.

A second concept uses the super-RENS (super-resolution near-field structure) effect[57–59]. This phenomenon allows the creation and reading of feature sizes considerably below the size of the laser wavelength used. In this case a masking layer is used above the storage layer. Making use of a nonlinear response of the masking layer, a sub-wavelength hole is formed in this layer. The precise origin of the super-RENS phenomenon has not yet been identified. In the past decade a range of rather different materials have been discovered that lead to a super-RENS effect when used as the masking layer[57–59]. Because the precise nature of the effect is not yet certain, it is difficult to predict the smallest feature size that can be reached.

One important issue for rewriteable storage solutions is the need to realize reversible phase-change recording, as the reading laser power is too high and erases the amorphous regions. Therefore alloys with higher transition temperatures are required. In the final section, we discuss a different storage concept that makes use of the unique property combination of phase-change alloys.

NON-VOLATILE ELECTRONIC MEMORIES

In the past decade, the potential of phase-change alloys for non-volatile electronic data storage has been discussed intensely. The concept has been sketched in Fig. 1. A short current pulse is used to crystallize an amorphous region locally. Because the crystalline region has a much higher electrical conductivity, reading of the information stored is fairly easy. In many phase-change alloys, the conductivity increases by more than three orders of magnitude, ensuring a high signal-to-noise ratio[45]. Local melting by a short, intense current pulse and subsequent quenching of the heated region erases the information. This scheme closely resembles the principle used in rewriteable optical data storage. Interestingly, the first materials to show phase-change properties were identified in studies of electrical switching behaviour[4] as early as 1968. These alloys were characterized by relatively slow crystallization speeds; hence fast crystallization was not feasible.

The advances in rewriteable optical storage made possible by the identification of superior materials with extremely short crystallization times have also led to a renaissance of non-volatile electronic memories. Nowadays large semiconductor companies such as Samsung[60], Intel[61], IBM[44], Macronix[44], STMicroelectronics[62], Qimonda[44], NXP[63], Hitachi[64], Renesas[64] and many others are actively investigating non-volatile memories based on phase-change materials. The phase-change alloys discovered in the past two decades have such fast crystallization kinetics that we can envisage their use in non-volatile memories that can compete in speed with dynamic random access memories (DRAM)[5,44,60,63,65–67]. This raises the hope that a universal memory can be developed that combines the simplicity of a DRAM with the speed of a static random access memory (SRAM) and the non-volatility of Flash memory[60,63,65]. Much of the insight obtained in the course of optimizing materials for rewriteable optical storage media can be re-used to tailor phase-change alloys for electronic storage. There are a number of specific points, however, that have not been addressed in optical storage media.

This can best be discussed with reference to Fig. 8, which shows the response of a phase-change memory cell in its amorphous state to an applied voltage. Initially for small applied voltages only a marginal current flows through the cell. This is due to the high resistivity of the amorphous state and leads to rather low heat dissipation in the cell, which would make recrystallization at low voltages practically impossible. Fortunately at a very moderate voltage the amorphous material undergoes a fast electronic transition (threshold switching), which leads to a much lower resistance in the amorphous state. A much larger current now flows through the amorphous region of the cell, producing enough heating to recrystallize the material. Hence it is important not only to understand the details of the mechanism involved in threshold switching[68] but also to know which materials show this switching mechanism. As mentioned in the introduction, all solids show a pronounced change of electrical resistance between the amorphous and the crystalline states. For applications as non-volatile memories, though, the useful materials are those in which recrystallization can be accomplished with lower power. Hence materials that show threshold switching are essential.

For non-volatile memories, mobile applications are particularly attractive, so materials are sought that can be switched with minimum power. Alloys with a relatively high resistance in the crystalline state and poor thermal conductivity are most promising in this respect. Furthermore, the whole aspect of integration of phase-change materials into semiconductor fabrication processes needs to be addressed. Given the stringent requirements for semiconductor performance, this creates considerable processing hurdles, which impede fast material optimization. At the same time this should be a strong motivation for fundamental research in the design of phase-change materials for non-volatile electronic memories.

We close with a few of the key questions that can guide in identifying materials with optimum properties. How can the power necessary to melt the crystalline region be minimized? Is reduction of the thermal conduction of the crystalline state the best choice or is it more efficient to identify materials with a low electrical conductivity? Are there phase-change alloys that are suitable for electrical memories that are incompatible with the requirements for optical storage? What is the underlying mechanism of threshold switching? Do redox-based memories[69] where conducting filaments are formed[63] have a common microscopic origin with phase-change alloys? Will the bottom-up approach recently suggested to revolutionize nanoelectronics hold promise for phase-change memories[70], that is, will phase-change wires deliever the ultimatte solution to scaling challenges? Although these questions are not meant as a complete list of challenges ahead, they are intended to highlight some of the opportunities for the design and optimization of phase-change materials in future applications that make use of their unique property portfolio.

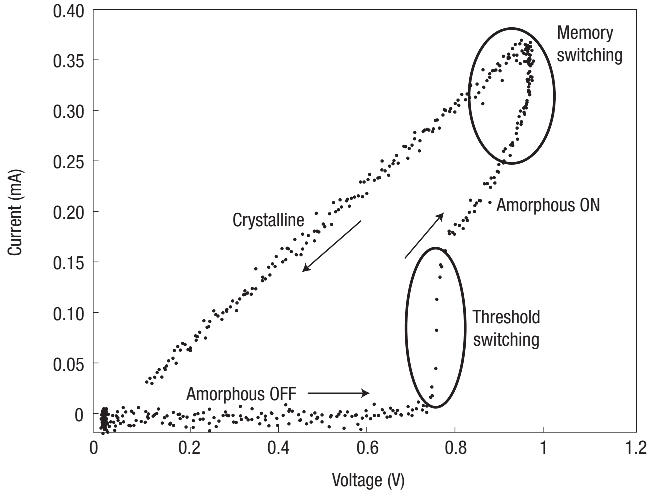

Figure 8 Typical current–voltage curve of a phase-change alloy that was initially in the amorphous state. On applying a small voltage only a marginal current flows initially, owing to the high resistance in the amorphous state. Above 0.7 V, the resistance drops and the current therefore increases greatly. Now heat is dissipated in the amorphous state, leading to the formation of the crystalline phase, which is characterized by a much lower resistance and hence high current.

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References

1. Meinders, E. R., Mijritskii, A. V., van Pieterson, L. & Wuttig, M. *Optical Data Storage: Phase Change Media and Recording* (Springer, Berlin, 2006).
2. Yamada, N. *et al.* High-speed overwritable phase-change optical disk material. *Jpn. J. Appl. Phys. Part 1* **26,** 61–66 (1987).
3. Satoh, I. & Yamada, N. DVD-RAM for all audio/video, PC, and network applications. *Proc. SPIE* **4085,** 283–290 (2001).
4. Ovshinsky, S. R. Reversible electrical switching phenomena in disordered structures. *Phys. Rev. Lett.* **21,** 1450–1453 (1968).
5. Wuttig, M. Phase-change materials—towards a universal memory? *Nature Mater.* **4,** 265–266 (2005).
6. Luo, M. B. & Wuttig, M. The dependence of crystal structure of Te-based phase-change materials on the number of valence electrons. *Adv. Mater.* **16,** 439–443 (2004).
7. Chen, M., Rubin, K. A. & Barton, R. W. Compound materials for reversible, phase-change optical-data storage. *Appl. Phys. Lett.* **49,** 502–504 (1986).
8. Yamada, N., Takenaga, M. & Takao, M. Te–Ge–Sn–Au phase-change recording film for optical disk. *Proc. SPIE* **695,** 79–85 (1986).
9. Ohno, E., Yamada, N., Kurumizawa, T., Kimura, K. & Takao, M. Tegesnau alloys for phase-change type optical disk memories. *Jpn. J. Appl. Phys. Part 1* **28,** 1235–1240 (1989).
10. Yamada, N., Ohno, E., Nishiuchi, K., Akahira, N. & Takao, M. Rapid-phase transitions of GeTe–Sb_2Te_3 pseudobinary amorphous thin-films for an optical disk memory. *J. Appl. Phys.* **69,** 2849–2856 (1991).
11. Kojima, R. *et al.* Nitrogen doping effect on phase change optical disks. *Jpn. J. Appl. Phys. Part 1* **37,** 2098–2103 (1998).
12. Kojima, R. & Yamada, N. Acceleration of crystallization speed by Sn addition to Ge–Sb–Te phase-change recording material. *Jpn. J. Appl. Phys. Part 1* **40,** 5930–5937 (2001).
13. Yusu, K., Nakai, T., Ashida, S., Ohmachi, N., Morishita, N. & Nakamura, N. Highspeed crystallization characteristics of Ge–Sb–Te–Bi materials used for next generation rewritable DVD with blue laser and NA = 0.65. *Proc. E\PCOS05* (2005); available at <http://www.epcos.org>.
14. Kusada, H., Hosaka, T., Kojima R. & Yamada, N. Effect of excess Sb on GeTe–Sb_2Te_3–Bi_2Te_3 recording films. *Proc. 18th Symp. PCOS2005* 32–35 (2006).
15. Iwasaki, H., Ide, Y., Harigaya, M., Kageyama, Y. & Fujimura, I. Completely erasable phase-change optical disk. *Jpn. J. Appl. Phys. Part 1* **31,** 461–465 (1992).
16. Horie, M., Nobukuni, N., Kiyono, K. & Ohno, T. High-speed rewritable DVD up to 20 m/s with nucleation-free eutectic phase-change material of Ge($Sb_{70}Te_{30}$)+Sb. *Proc. SPIE* **4090,** 135–143 (2000).
17. Kato, T. *et al.* The phase change optical disc with the data recording rate of 140 Mbps. *Jpn. J. Appl. Phys. Part 1* **41,** 1664–1667 (2002).
18. Iwasaki, H. *et al.* Completely erasable phase-change optical disc. II. Application of Ag–In–Sb–Te mixed-phase system for rewritable compact disc compatible with CD-velocity and double CD-velocity. *Jpn. J. Appl. Phys. Part 1* **32,** 5241–5247 (1993).

19. Afonso, C. N., Solis, J., Catalina, F. & Kalpouzos, C. Ultrafast reversible phase-change in GeSb films for erasable optical storage. *Appl. Phys. Lett.* **60,** 3123–3125 (1992).
20. Yuzurihara, H., Iwasa, H. & Kaneko, Y. GeSbSnMn for high speed BD-RE media. *Proc. 17th Symp. PCOS2005* 19–22 (2005).
21. Yamada, N. & Matsunaga, T. Structure of laser-crystallized $Ge_2Sb_{2+x}Te_5$ sputtered thin films for use in optical memory. *J. Appl. Phys.* **88,** 7020–7028 (2000).
22. Matsunaga, T. & Yamada, N. A study of highly symmetrical crystal structures, commonly seen in high-speed phase-change materials, using synchrotron radiation. *Jpn. J. Appl. Phys. Part 1* **41,** 1674–1678 (2002).
23. Matsunaga, T. & Yamada, N. Crystallographic studies on high-speed phase-change materials used for rewritable optical recording disks. *Jpn. J. Appl. Phys. Part 1* **43,** 4704–4712 (2004).
24. Nonaka, T., Ohbayashi, G., Toriumi, Y., Mori, Y. & Hashimoto, H. Crystal structure of GeTe and $Ge_2Sb_2Te_5$ meta-stable phase. *Thin Solid Films* **370,** 258–261 (2000).
25. Matsunaga, T. & Yamada, N. Crystal structure and bonding nature of $Ge_8Sb_2Te_{11}$, a suitable material for high-speed, high density phase-change recording. *Proc. 16th Symp. PCOS2004* 1–4 (2005).
26. Kolobov, A. V. *et al.* Understanding the phase-change mechanism of rewritable optical media. *Nature Mater.* **3,** 703–708 (2004).
27. Matsunaga, T., Umetani, Y. & Yamada, N. Structural study of a $Ag_{3.4}In_{3.7}Sb_{76.4}Te_{16.5}$ quadruple compound utilized for phase-change optical disks. *Phys. Rev. B* **64,** 184116 (2001).
28. Zallen, R. Models of amorphous solids. *J. Non-Cryst. Solids* **75,** 3–14 (1985).
29. Zachariasen, W. The atomic arrangement in glass. *J. Am. Chem. Soc.* **54,** 3841–3851 (1932).
30. Thorpe, M. F. Continuous deformations in random networks. *J. Non-Cryst. Solids* **57,** 355–370 (1983).
31. Phillips, J. C. & Thorpe, M. F. Constraint theory, vector percolation and glass-formation. *Solid State Commun.* **53,** 699–702 (1985).
32. Feng, X. W., Bresser, W. J. & Boolchand, P. Direct evidence for stiffness threshold in chalcogenide glasses. *Phys. Rev. Lett.* **78,** 4422–4425 (1997).
33. Welnic, W. *et al.* Unravelling the interplay of local structure and physical properties in phase-change materials. *Nature Mater.* **5,** 56–62 (2006).
34. Welnic, W., Botti, S., Reining, L. & Wuttig, M. Origin of the optical contrast in phase change materials. *Phys. Rev. Lett.* **98,** 236403 (2007).
35. Debenedetti, P. G. & Stillinger, F. H. Supercooled liquids and the glass transition. *Nature* **410,** 259–267 (2001).
36. Baker, D. A., Paesler, M. A., Lucovsky, G., Agarwal, S. C. & Taylor, P. C. Application of bond constraint theory to the switchable optical memory material $Ge_2Sb_2Te_5$. *Phys. Rev. Lett.* **96,** 255501 (2006).
37. Kohara, S. *et al.* Structural basis for the fast phase change of $Ge_2Sb_2Te_5$: ring statistics analogy between the crystal and amorphous states. *Appl. Phys. Lett.* **89,** 201910 (2006).
38. Kalb, J., Spaepen, F. & Wuttig, M. Calorimetric measurements of phase transformations in thin films of amorphous Te alloys used for optical data storage. *J. Appl. Phys.* **93,** 2389–2393 (2003).
39. Kalb, J., Spaepen, F. & Wuttig, M. Atomic force microscopy measurements of crystal nucleation and growth rates in thin films of amorphous Te alloys. *Appl. Phys. Lett.* **84,** 5240–5242 (2004).
40. Kalb, J. A., Wuttig, M. & Spaepen, F. Calorimetric measurements of structural relaxation and glass transition temperatures in sputtered films of amorphous Te alloys used for phase change recording. *J. Mater. Res.* **22,** 748–754 (2007).
41. Turnbull, D. Under what conditions can a glass be formed? *Contemp. Phys.* **10,** 473–488 (1969).
42. Kalb, J. A., Spaepen, F. & Wuttig, M. Kinetics of crystal nucleation in undercooled droplets of Sb- and Te-based alloys used for phase change recording. *J. Appl. Phys.* **98,** 054910 (2005).
43. Friedrich, I., Weidenhof, V., Lenk, S. & Wuttig, M. Morphology and structure of laser-modified $Ge_2Sb_2Te_5$ films studied by transmission electron microscopy. *Thin Solid Films* **389,** 239–244 (2001).
44. Chen, Y. C. *et al.* Ultra-thin phase-change bridge memory device using GeSb. *IEDM Tech. Digest* 777–780 (2006).
45. Friedrich, I., Weidenhof, V., Njoroge, W., Franz, P. & Wuttig, M. Structural transformations of $Ge_2Sb_2Te_5$ films studied by electrical resistance measurements. *J. Appl. Phys.* **87,** 4130–4134 (2000).
46. Coombs, J. H., Jongenelis, A. P. J. M., Vanesspiekman, W. & Jacobs, B. A. J. Laser-induced crystallization phenomena in GeTe-based alloys. 2. Composition dependence of nucleation and growth. *J. Appl. Phys.* **78,** 4918–4928 (1995).
47. Weidenhof, V., Pirch, N., Friedrich, I., Ziegler, S. & Wuttig, M. Minimum time for laser induced amorphization of $Ge_2Sb_2Te_5$ films. *J. Appl. Phys.* **88,** 657–664 (2000).
48. Anderson, P. W. Absence of diffusion in certain random lattices. *Phys. Rev.* **109,** 1492–1505 (1958).
49. Njoroge, W. K., Woltgens, H. W. & Wuttig, M. Density changes upon crystallization of $Ge_2Sb_{2.04}Te_{4.74}$ films. *J. Vac. Sci. Technol. A* **20,** 230–233 (2002).
50. Yamada, N. *et al.* Phase-change material for use in rewritable dual-layer optical disk. *Proc. SPIE* **4342,** 55–63 (2002).
51. Onida, G., Reining, L. & Rubio, A. Electronic excitations: density-functional versus many-body Green's-function approaches. *Rev. Mod. Phys.* **74,** 601–659 (2002).
52. Stuke, J. & Zimmerer, G. Optical properties of amorphous 3–5 compounds.1. Experiment. *Phys. Status Solidi B* **49,** 513–523 (1972).
53. Wuttig, M. *et al.* The role of vacancies and local distortions in the design of new phase-change materials. *Nature Mater.* **6,** 122–128 (2007).
54. Peierls, R. E. *Quantum Theory of Solids* (Oxford Univ. Press, Oxford, 1956).
55. Gaspard, J. P. Hume–Rothery rule in V–VI compounds. *Solid State Commun.* **84,** 839–842 (1992).
56. Gaspard, J. P., Pellegatti, A., Marinelli, F. & Bichara, C. Peierls instabilities in covalent structures. I. Electronic structure, cohesion and the $Z = 8 - N$ rule. *Phil. Mag. B* **77,** 727–744 (1998).
57. Nakano, T., Sato, A., Fuji, H., Tominaga, J. & Atoda, N. Transmitted signal detection of optical disks with a superresolution near-field structure. *Appl. Phys. Lett.* **75,** 151–153 (1999).
58. Tominaga, J., Fuji, H., Sato, A., Nakano, T. & Atoda, N. The characteristics and the potential of super resolution near-field structure. *Jpn. J. Appl. Phys. Part 1* **39,** 957–961 (2000).
59. Kim, J. *et al.* Super-resolution near-field structure with alternative recording and mask materials. *Jpn. J. Appl. Phys. Part 1* **42,** 1014–1017 (2003).
60. Cho, W. Y. *et al.* A 0.18-μm 3.0-V 64-Mb nonvolatile phase-transition random access memory (PRAM). *IEEE J. Solid-State Circuits* **40,** 293–300 (2005).
61. Hudgens, S. & Johnson, B. Overview of phase-change chalcogenide nonvolatile memory technology. *Mater. Res. Soc. Bull.* **29,** 829–832 (2004).
62. Bez, R. & Pirovano, A. Non-volatile memory technologies: emerging concepts and new materials, *Mater. Sci. Semicond. Proc.* **7,** 349–355 (2004).
63. Lankhorst, M. H. R., Ketelaars, B. W. S. M. M. & Wolters, R. A. M. Low-cost and nanoscale non-volatile memory concept for future silicon chips. *Nature Mater.* **4,** 347–352 (2005).
64. Hanzawa, S. *et al.* A 512 kB embedded phase change memory with 416 kb/s write throughput 100 μA cell write current. *ISSCC Digest Tech.* 474–475 (2007).
65. Pirovano, A., Lacaita, A. L., Benvenuti, A., Pellizzer, F. & Bez, R. Electronic switching in phase-change memories. *IEEE Trans. Electron Devices* **51,** 452–459 (2004).
66. Merget, F., Kim, D. H., Bolivar, P. H. & Kurz, H. Lateral phase change random access memory cell design for low power operation. *Microsyst. Technol.* **13,** 169–172 (2007).
67. Kim, D. H., Merget, F., Forst, M. & Kurz, H. Threedimensional simulation model of switching dynmics in phase change random access memory cells. *J. Appl. Phys.* **101,** 064512–1 (2007).
68. Kastner, M., Adler, D. & Fritsche, H. Valence-alternation model for localized gap states in lone-pair semiconductors. *Phys. Rev. Lett.* **37,** 1504–1507 (1976).
69. Waser, R. & Aono, M. Nanoionics-based resistive switching memories. *Nature Mater.* **6,** 833–840 (2007).
70. Lu, W. & Lieber, C. M. Nanoelectronics from the bottom up. *Nature Mater.* **6,** 841–850 (2007).

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Phase-change materials for rewriteable data storage

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In Fig. 2 of this Insight Review Article, the bottom label in each box was incorrect. The figure should have appeared as shown below.